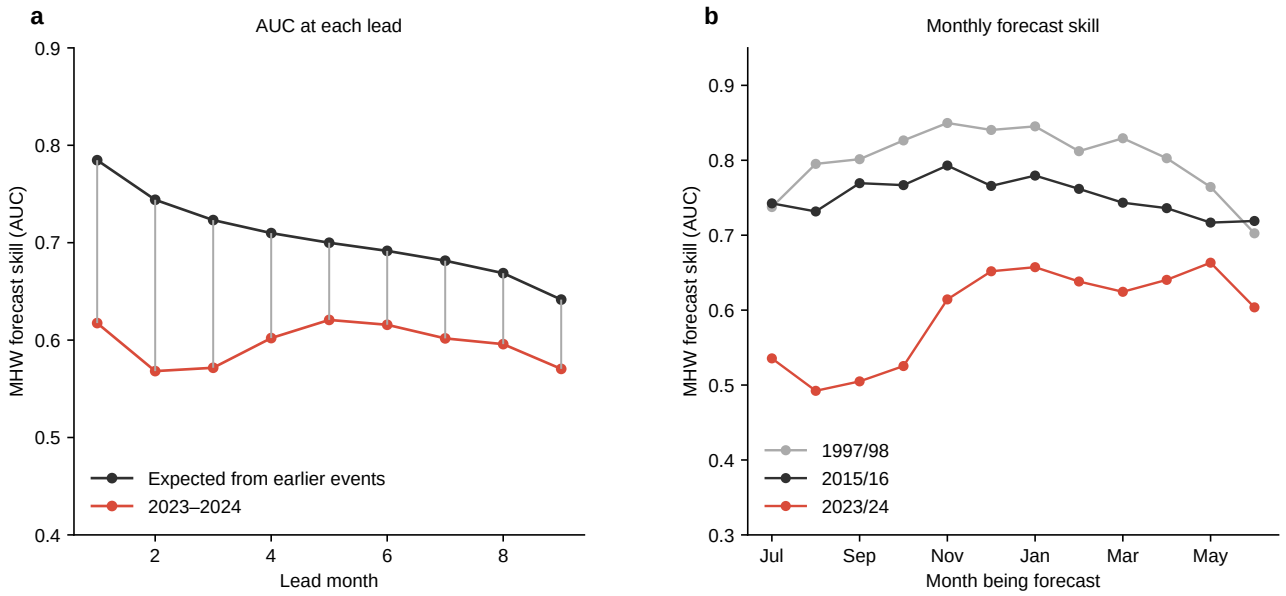
Supplementary Fig. 4 | MHW forecast skill across lead months and event development



a, AUC expected from earlier events (dark line) and realized in 2023–2024 (red line), evaluated separately at each lead from 1 to 9 months. Connecting vertical segments show the expected-minus-realized difference. At each lead, AUC is averaged over July–June and regressed on El Niño amplitude using the nine earlier events; 2023–2024 is excluded. b, AUC averaged over leads 1–9 for each verifying month in the three strong events. The light-grey, dark-grey and red lines denote 1997–1998, 2015–2016 and 2023–2024. All 1,080 event–month–lead records are used: ten events × twelve months × nine leads. Leads and months are repeated evaluations, not independent events.

No lead-specific significance test is claimed. These panels retain the forecast extraction and monthly spatial-AUC definition of main Fig. 1a,b.